

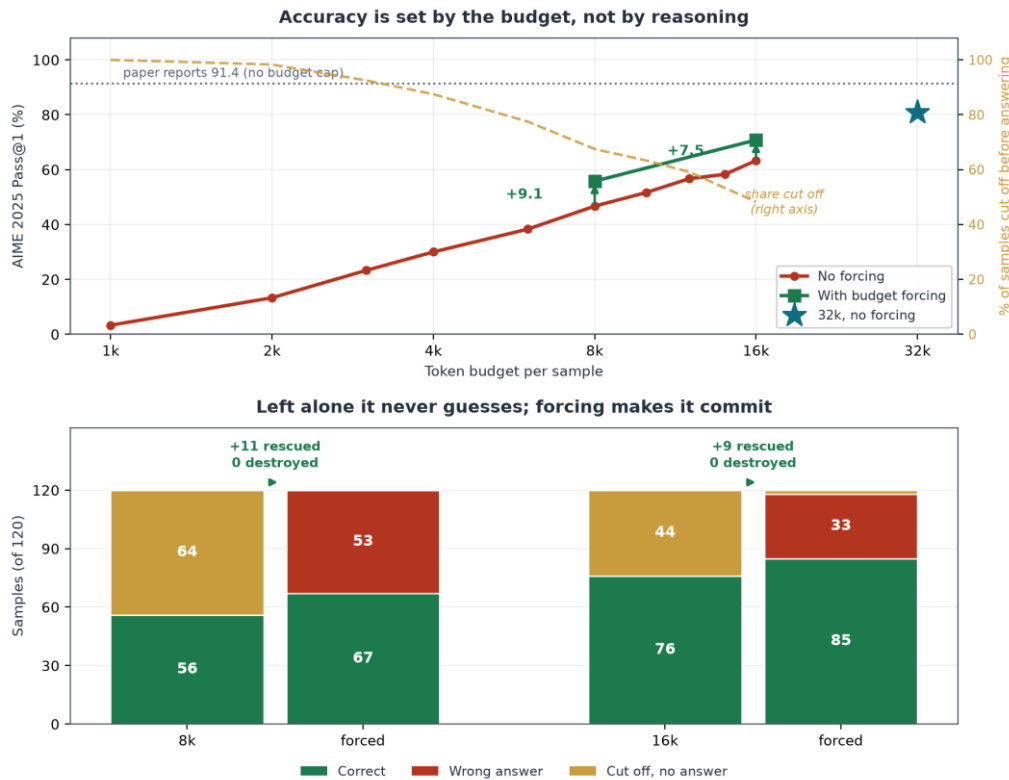
Reasoning Termination, Not Reasoning Capability

Budget forcing on VibeThinker-3B, AIME 2025

Abu Ahmad · 2121725042 · CSE 465 · North South University

The finding

VibeThinker-3B scores **91.4** on AIME 2025 in its paper; we reproduced **80.8**. The gap is not weaker reasoning — the model solves the problem and then never writes the answer down. Of 120 sampled attempts, essentially every one that finished was correct, while every one cut off by the token budget scored zero for want of a boxed answer. Across 1,200 sample-budget observations the model volunteered **exactly one wrong answer**: it states the correct answer or states nothing at all. Interrupting it at the budget and forcing it to commit recovers **7 to 13 points**, has **never** destroyed a correct answer, and costs about five tokens.



Top: accuracy rises with the token budget as truncation falls — mirror images. Bottom: forcing converts the amber “no answer” block into correct and wrong, and the green never shrinks.

Table 1 · Main result

Budget	No forcing	+ Forcing	Gain	McNemar p
4k	27.5%	40.8%	+13.3	—
8k	46.7%	55.8%	+9.1	0.0026
16k	63.3%	70.8%	+7.5	0.0077
32k	80.8%	not run	—	—

Table 2 · Does forcing ever hurt?

Budget	Truncated	Rescued	Damaged	p
8k	81	11	0	0.0026
16k	58	9	0	0.0077

Zero damage across 139 interventions. Forcing only ever touches a sample that had already failed.

Table 3 · What the 120 samples are made of

Condition	Correct	Wrong	No answer
8k, no forcing	56	0	64
8k, forced	67	53	0
16k, no forcing	76	0	44
16k, forced	85	33	2

The zero column is the finding: left alone, the model never guesses.